

Replay-Aware ATSP Benchmark Report

Overview

- Condition count: 4.
- Best final transfer gap: atsp_no_replay.
- Best TSPLIB ATSP holdout gap: atsp_no_replay.
- Best synthetic holdout gap: atsp_failure_replay.

Run Metadata

- run_name: run_20260513_171236_h
- started_at_local: 2026-05-13 17:12:36
- finished_at_local: 2026-05-13 17:21:04
- duration_hhmm: 00:08
- duration_seconds: 507.918
- seed_offset: 7000
- replicate_label: h
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final TSPLIB ATSP Gap	Final Synthetic Gap	Final Transfer Gap	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
atsp_no_replay	none	score_only	False	0.169705	0.030781	0.100243	0.0	0.78	0.0
atsp_random_replay	random	score_only	False	0.258493	0.037956	0.148224	0.0	0.78	0.0
atsp_failure_replay	failure	score_only	False	0.192846	0.009004	0.100925	0.0	0.58	0.0
atsp_failure_replay_compression	failure	novelty_gate	True	0.192846	0.009004	0.100925	0.858717	0.58	0.036479

Condition Notes

atsp_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: TSPLIB ATSP 0.169705, synthetic 0.030781, combined 0.100243.

- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.78.
- Adaptation efficiency 0.0 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.169705 across 4 instances; family means: ft=0.228675, ftv=0.292622, p=0.002135, ry=0.155388.
- Panel synthetic_holdout mean gap 0.030781 across 4 instances; family means: clockwise_ring=0.085613, corridor_drift=0.0, hub_spokes=0.037511, wind_clusters=0.0.
- Worst recent training cases: [{"best_known_cost": 1530, "cost": 1914, "family": "ftv", "name": "ftv38", "optimality_gap": 0.25098}, {"best_known_cost": 1286, "cost": 1491, "family": "ftv", "name": "ftv33", "optimality_gap": 0.159409}, {"best_known_cost": 1473, "cost": 1690, "family": "ftv", "name": "ftv35", "optimality_gap": 0.147318}]

atsp_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: TSPLIB ATSP 0.258493, synthetic 0.037956, combined 0.148224.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.78.
- Adaptation efficiency 0.0 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.258493 across 4 instances; family means: ft=0.447647, ftv=0.398016, p=0.012811, ry=0.175496.
- Panel synthetic_holdout mean gap 0.037956 across 4 instances; family means: clockwise_ring=0.085613, corridor_drift=0.0, hub_spokes=0.066212, wind_clusters=0.0.
- Worst recent training cases: [{"best_known_cost": 1286, "cost": 1574, "family": "ftv", "name": "ftv33", "optimality_gap": 0.22395}, {"best_known_cost": 1530, "cost": 1827, "family": "ftv", "name": "ftv38", "optimality_gap": 0.194118}, {"best_known_cost": 1473, "cost": 1722, "family": "ftv", "name": "ftv35", "optimality_gap": 0.169043}]

atsp_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: TSPLIB ATSP 0.192846, synthetic 0.009004, combined 0.100925.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.58.
- Adaptation efficiency 0.0 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.192846 across 4 instances; family means: ft=0.319768, ftv=0.33478, p=0.004093, ry=0.112744.
- Panel synthetic_holdout mean gap 0.009004 across 4 instances; family means: clockwise_ring=0.0, corridor_drift=0.0, hub_spokes=0.033816, wind_clusters=0.002199.
- Worst recent training cases: [{"best_known_cost": 1530, "cost": 1939, "family": "ftv", "name": "ftv38", "optimality_gap": 0.26732}, {"best_known_cost": 1286, "cost": 1537, "family": "ftv", "name": "ftv33", "optimality_gap": 0.195179}, {"best_known_cost": 1473, "cost": 1705, "family": "ftv", "name": "ftv35", "optimality_gap": 0.157502}]

atsp_failure_replay_compression

- Replay mode: failure with selection mode novelty_gate.
- Final transfer gaps: TSPLIB ATSP 0.192846, synthetic 0.009004, combined 0.100925.
- Accepted-epoch count 2, mean accepted code novelty 0.858717, and final complexity 0.58.
- Adaptation efficiency 0.036479 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.

- Panel heldout_tsplib mean gap 0.192846 across 4 instances; family means: ft=0.319768, ftv=0.33478, p=0.004093, ry=0.112744.
- Panel synthetic_holdout mean gap 0.009004 across 4 instances; family means: clockwise_ring=0.0, corridor_drift=0.0, hub_spokes=0.033816, wind_clusters=0.002199.
- Worst recent training cases: [{"best_known_cost": 1530, "cost": 1939, "family": "ftv", "name": "ftv38", "optimality_gap": 0.26732}, {"best_known_cost": 1473, "cost": 1705, "family": "ftv", "name": "ftv35", "optimality_gap": 0.157502}, {"best_known_cost": 1286, "cost": 1478, "family": "ftv", "name": "ftv33", "optimality_gap": 0.1493}]

Judge Appendix

Summary of Results for Replay-Aware ATSP Benchmark Suite

Condition	Synth etic Gap	TSP LIB Gap	Tran sfer Gap	Replay Mode	Co mpl exit y	Code Nove lty	Notes on Transfer and Novelty
atsp_no_replay (best holdout & transfer)	0.0308	0.1697	0.1002	none	0.78	0.0	Best transfer to held-out TSPLIB and synthetic; no code novelty observed.
atsp_failure_replay (best synthetic)	0.0090	0.1928	0.1009	failure	0.58	0.0	Best synthetic gap (holdout synthetic close to zero); transfer gap slightly worse than no_replay.
atsp_failure_replay_compression	0.0090	0.1928	0.1009	failure + compression	0.58	0.86	Same transfer and gaps as failure_replay but elevated code novelty and compression pressure present.
atsp_random_replay	0.0380	0.2585	0.1482	random	0.78	0.0	Worst transfer and TSPLIB gaps; no code novelty observed.

Interpretation

- Held-Out TSPLIB ATSP Gap & Transfer Evidence (Primary Metrics):**
 - The **no replay** (atsp_no_replay) condition yields the **lowest mean TSPLIB gap (0.1697)** and best transfer gap (0.1002), indicating superior generalization to real-world TSPLIB instances.
 - Both **failure replay conditions** (with and without compression) have slightly higher TSPLIB gaps (~0.193) and similar transfer gaps (~0.101), slightly worse than no replay.
 - Random replay** performs worst on TSPLIB (0.2585) and transfer gaps (0.1482), showing degradation from replaying indiscriminately.
- Synthetic Holdout Gap:**
 - Failure replay (both standard and compression-aware) achieves the lowest synthetic holdout gaps (~0.009), substantially better than no replay (0.031) or random replay (0.038), demonstrating clear synthetic domain improvement with failure replay.
 - This suggests failure replay promotes more specialized adaption to synthetic distributions.
- Code Novelty vs. Transfer:**
 - No replay, random replay, and failure replay (standard) conditions have zero code novelty.**

- **Failure replay compression condition exhibits high code novelty (mean ~0.86).** However, this does not improve transfer or TSPLIB performance over standard failure replay.
- Hence, **increased code novelty due to compression pressure does not translate into better transfer** metrics.
- This indicates that lexical/code novelty here is not synonymous with algorithmic innovation in transfer performance.

4. **Replay Mode Effects:**

- **No replay yields best transfer and TSPLIB performance, despite zero adaptation efficiency and code novelty.**
- **Failure replay improves synthetic holdout gap strongly, with comparable transfer to no replay, but no code novelty.**
- **Random replay harms transfer and TSPLIB gaps, suggesting naive replay can be detrimental.**
- **Compression-aware failure replay increases code novelty and adaptation efficiency slightly but does not improve transfer or TSPLIB gap relative to failure replay without compression.**

5. **Complexity:**

- Failure replay (with/without compression) conditions have lower mean complexity (0.58) compared to no replay and random replay (0.78).
- Lower complexity with similar or slightly worse TSPLIB gaps may indicate more streamlined strategies under failure replay.

Conclusions

- **For transfer (TSPLIB and synthetic holdout), no replay yields the best held-out transfer gap and TSPLIB gap;** thus, replay is not strictly necessary or beneficial for transfer in this benchmark.
- **Failure replay significantly improves synthetic holdout gaps but does not improve (may slightly degrade) transfer on held-out TSPLIB problems compared to no replay.**
- **Random replay consistently harms performance.**
- **Compression-aware failure replay increases code novelty substantially but without corresponding transfer gains over failure replay without compression, indicating novelty here is not algorithmically advantageous transfer-wise.**
- Overall, prioritize **no replay** for best transfer generalization and **failure replay** for synthetic holdout accuracy; be cautious of indiscriminate or random replay harms.
- Lexical/code novelty increases under compression-aware replay, but this is not supported by deterministic transfer metrics as algorithmic invention.

Recommendation:

Focus on no replay or targeted failure replay adaptations for robust transfer in ATSP benchmarks, and treat code novelty gains under compression with skepticism unless accompanied by lowered transfer or TSPLIB gaps.